

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – VIII

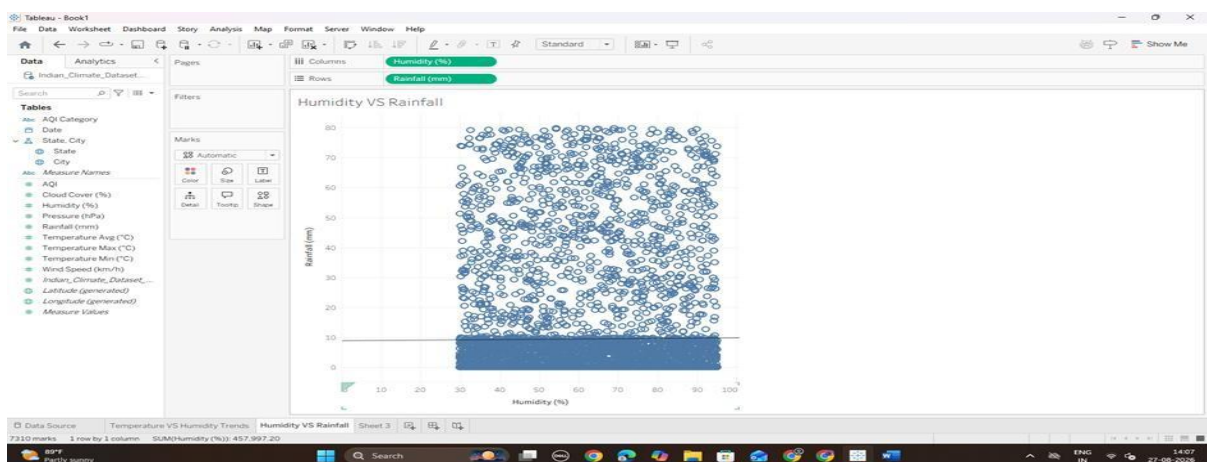
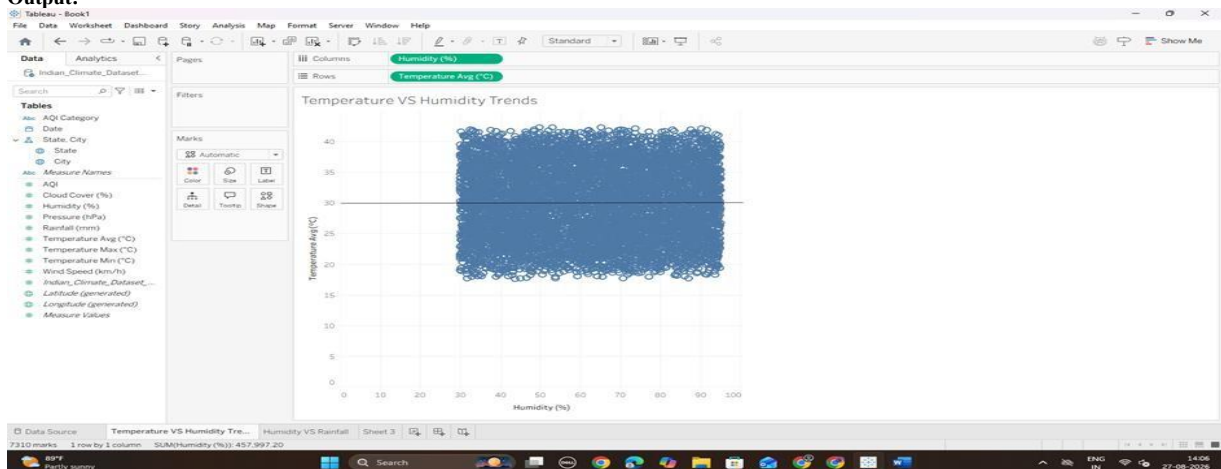
Roll No.: T001	Name: Soham Acharekar
Class: TYIT	Batch: 01
Date of Assignment: 27/08/2026	Date/Time of Submission: 27/08/2026

a) Add and Interpret Trend Lines (Linear and Exponential)

Steps:

1. Connect **Indian_Climate_Dataset_2024_2025.csv** in Tableau.
2. Create a worksheet named **Temperature vs Humidity Trends**.
3. Drag **Humidity (%)** → **Columns** and **Temperature_Avg (°C)** → **Rows**.
4. Uncheck **Analysis** → **Aggregate Measures**.
5. Open the Analytics pane.
6. Drag **Trend Line** onto the chart and select **Linear**.
7. Hover over the trend line to view the equation, R^2 and p-value.
8. Create **Humidity vs Rainfall** worksheet.
9. Drag **Humidity (%)** → **Columns** and **Rainfall (mm)** → **Rows**.
10. Add an **Exponential Trend Line**.

Output:

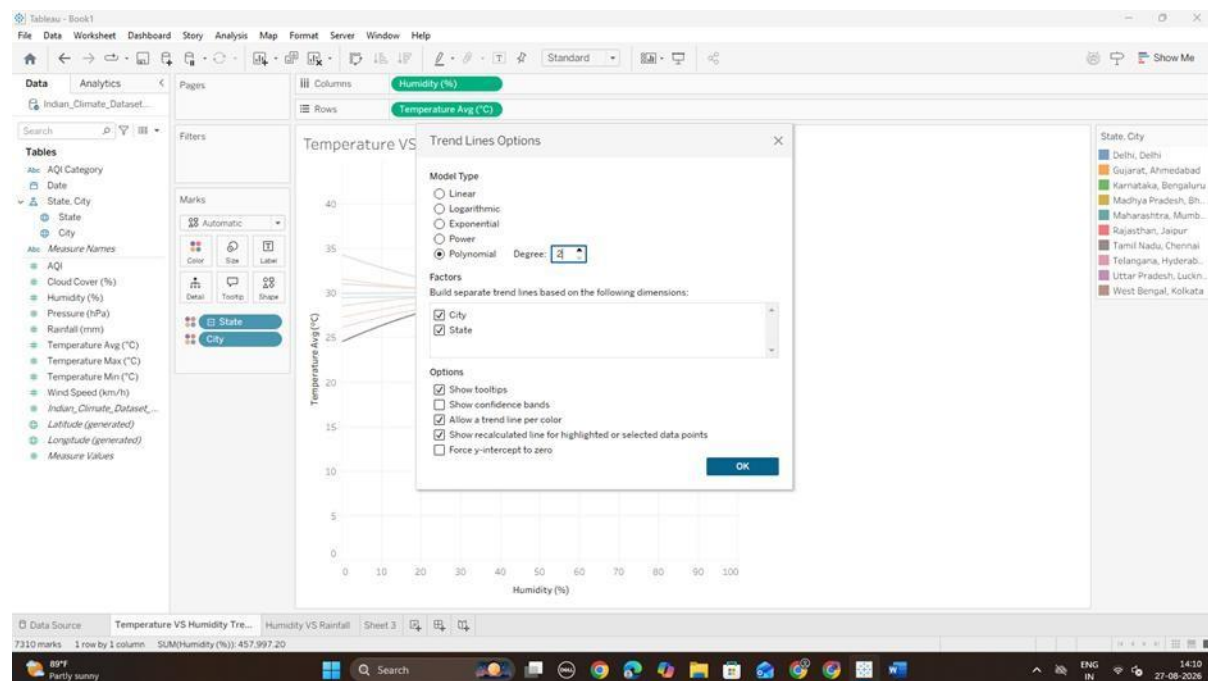
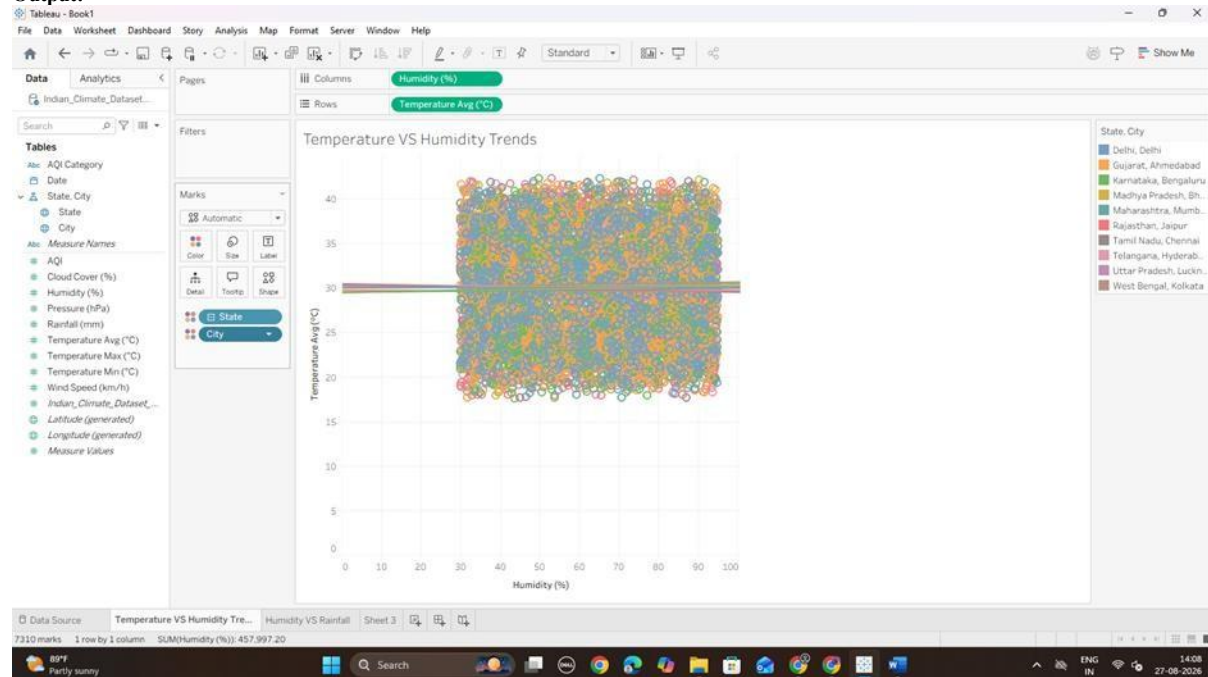


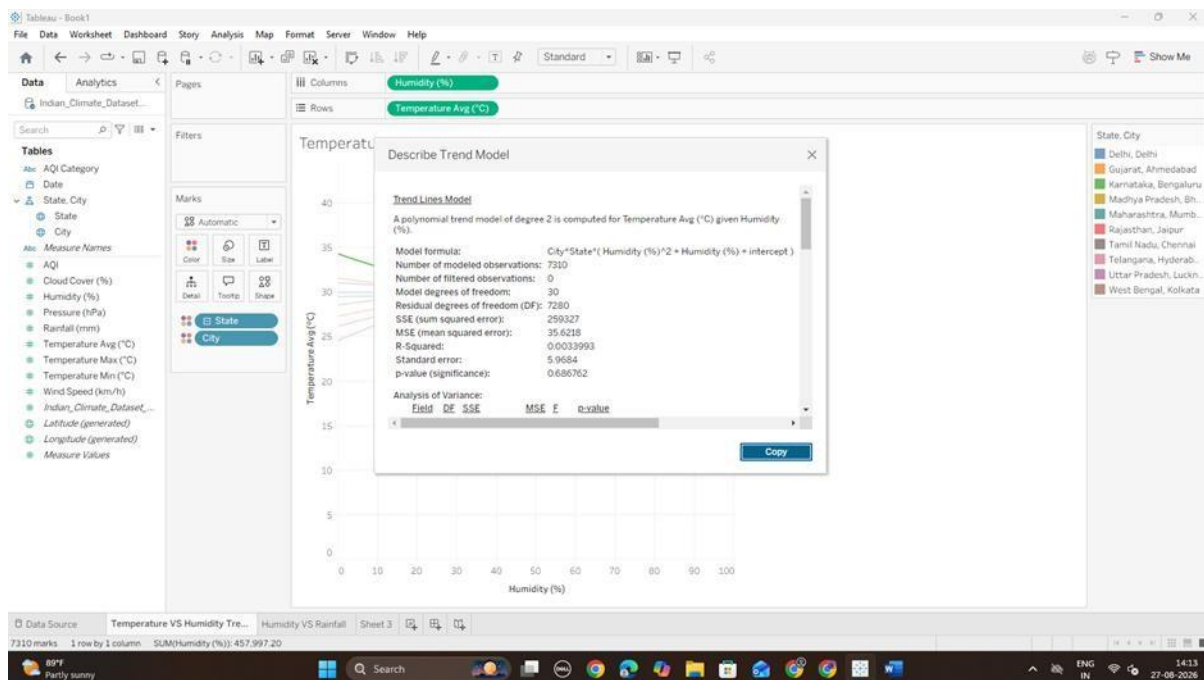
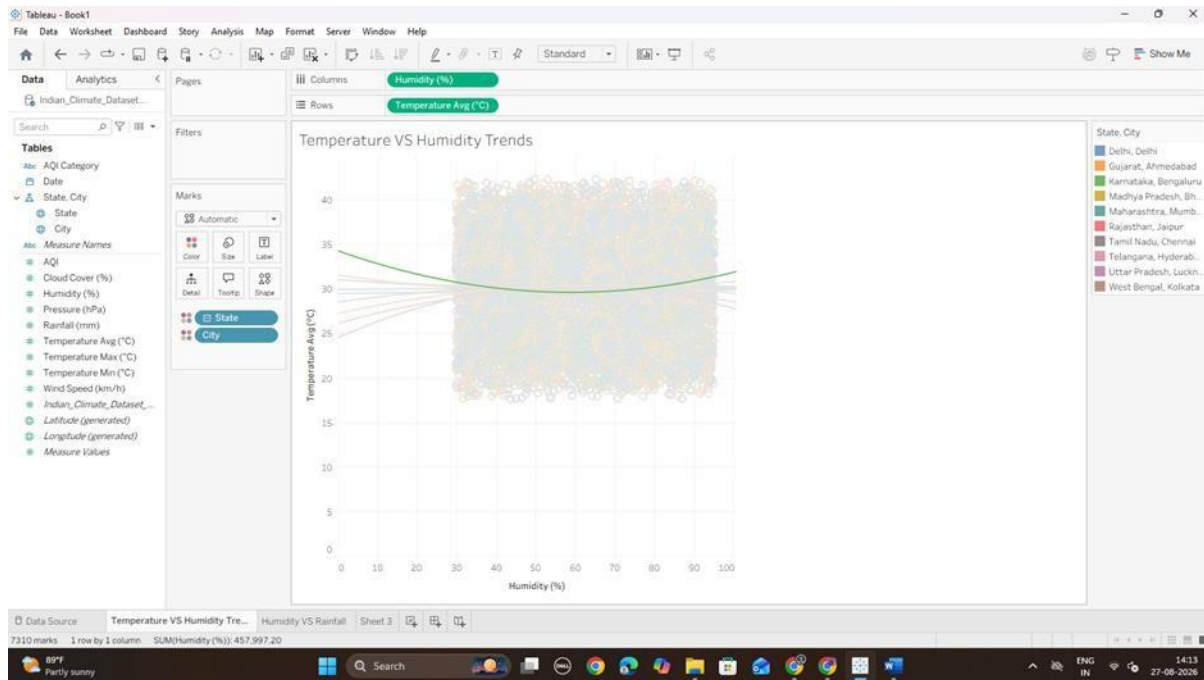
b) Customize and Analyze Trend Models

Steps:

1. Open Temperature vs Humidity Trends.
2. Drag **City** → **Color** on the Marks card.
3. Observe separate trend lines for different cities.
4. Right-click the trend line and select Edit Trend Lines.
5. Compare Linear, Logarithmic, Exponential and Polynomial models.
6. Select Polynomial (Degree 2) and observe the result.
7. Right-click the chart and select Describe Trend Line.
8. Select Describe Trend Model to view ANOVA, Degrees of Freedom, SSE and F-statistic.
9. Compare the R^2 values for different cities/models.

Output:

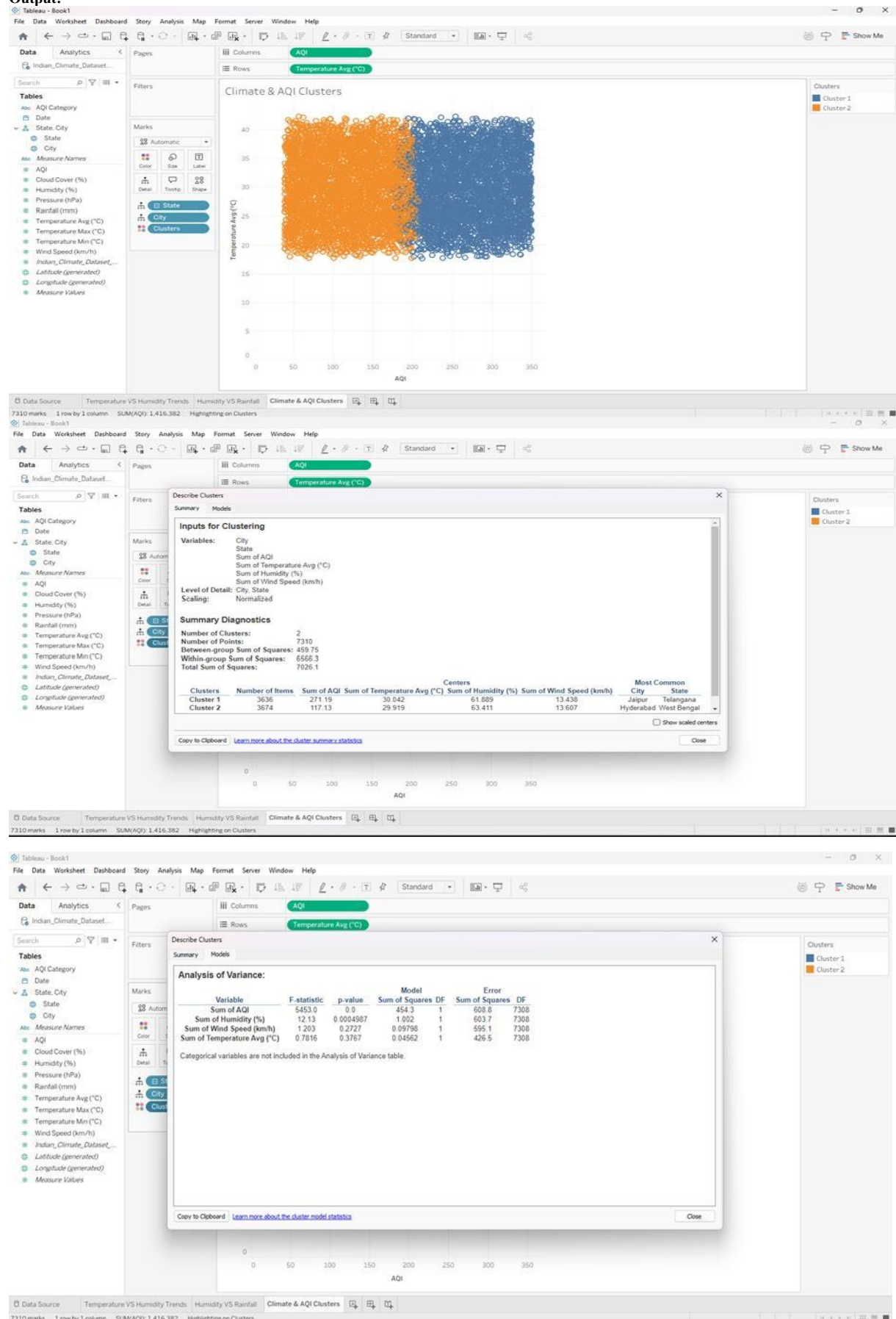




c) Perform Clustering on Datasets

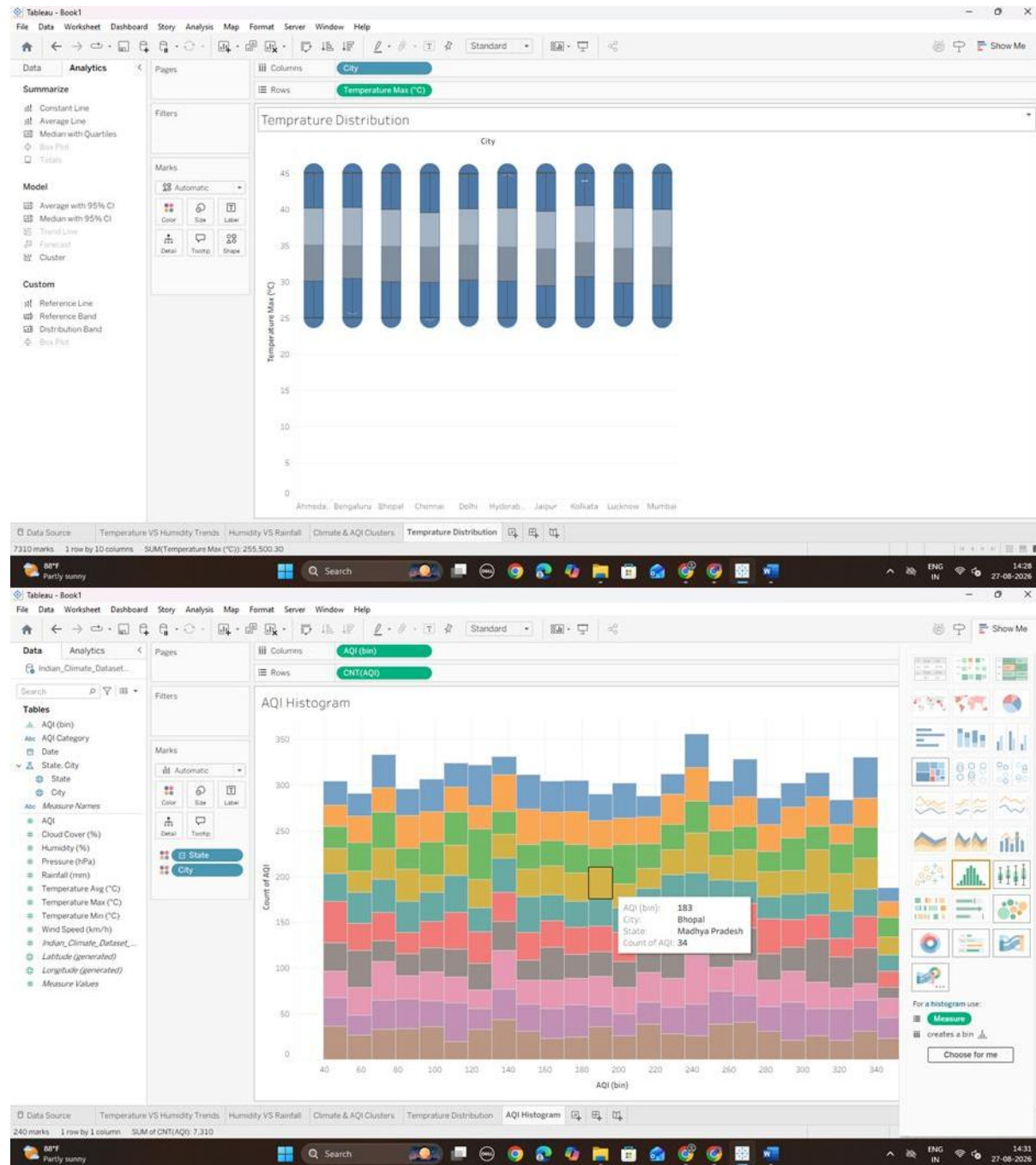
Steps:

1. Create a worksheet named Climate & AQI Clusters.
2. Drag **AQI** → **Columns** and **Temperature_Avg (°C)** → **Rows**.
3. Uncheck **Analysis** → **Aggregate Measures**.
4. Drag **City** → **Detail** on the Marks card.
5. Open the Analytics pane.
6. Drag **Cluster** onto the chart.
7. Add **AQI**, **Temperature_Avg (°C)**, **Humidity (%)** and **Wind_Speed (km/h)** as variables.
8. Keep the number of clusters Automatic, or select 3 or 4.
9. Select **Describe Clusters** to view cluster summaries and model statistics.

Output:**d) Analyze Distributions Using Built-in Tools**

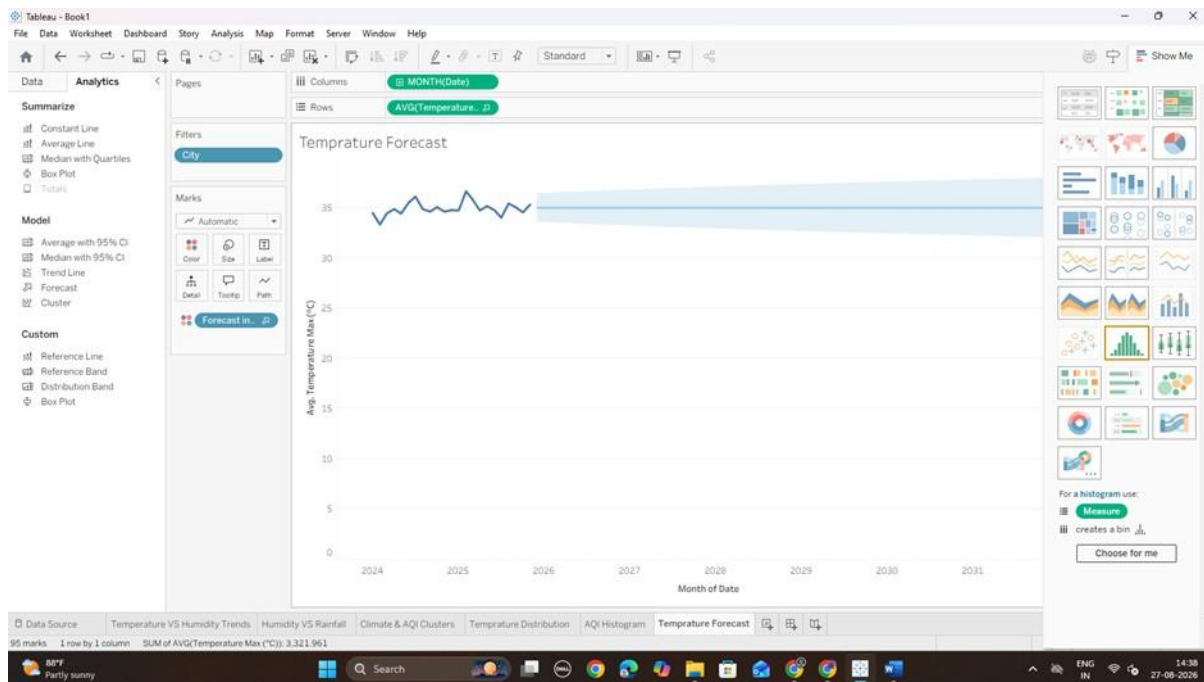
Steps:

1. Create a worksheet named Temperature Distributions.
2. Drag **City** → **Columns** and **Temperature_Max (°C)** → **Rows**.
3. Open the Analytics pane.
4. Drag Box Plot onto the chart.
5. Observe median, quartiles, whiskers and outliers.
6. Create another worksheet named AQI Histogram.
7. Select **AQI** and open Show Me.
8. Select the Histogram option.
9. Drag **City** → **Color** to compare AQI distributions.

Output:**e) Apply Forecasting Techniques**

Steps:

1. Create a worksheet named Temperature Forecast.
2. Drag **Date** → **Columns**.
3. Change Date to Continuous Month.
4. Drag **Temperature_Max (°C)** → **Rows** and use AVG (Temperature_Max).
5. Drag **City** → **Filters** and select 2–3 cities.
6. Open the Analytics pane.
7. Drag Forecast onto the chart.
8. Open **Forecast** → **Forecast Options**.
9. Set Forecast Length = Exactly 6 Months.
10. Set Prediction Interval = 95%.
11. Select Describe Forecast to review RMSE and the selected model.

Output:**f) Compare Different Statistical Models****Steps:**

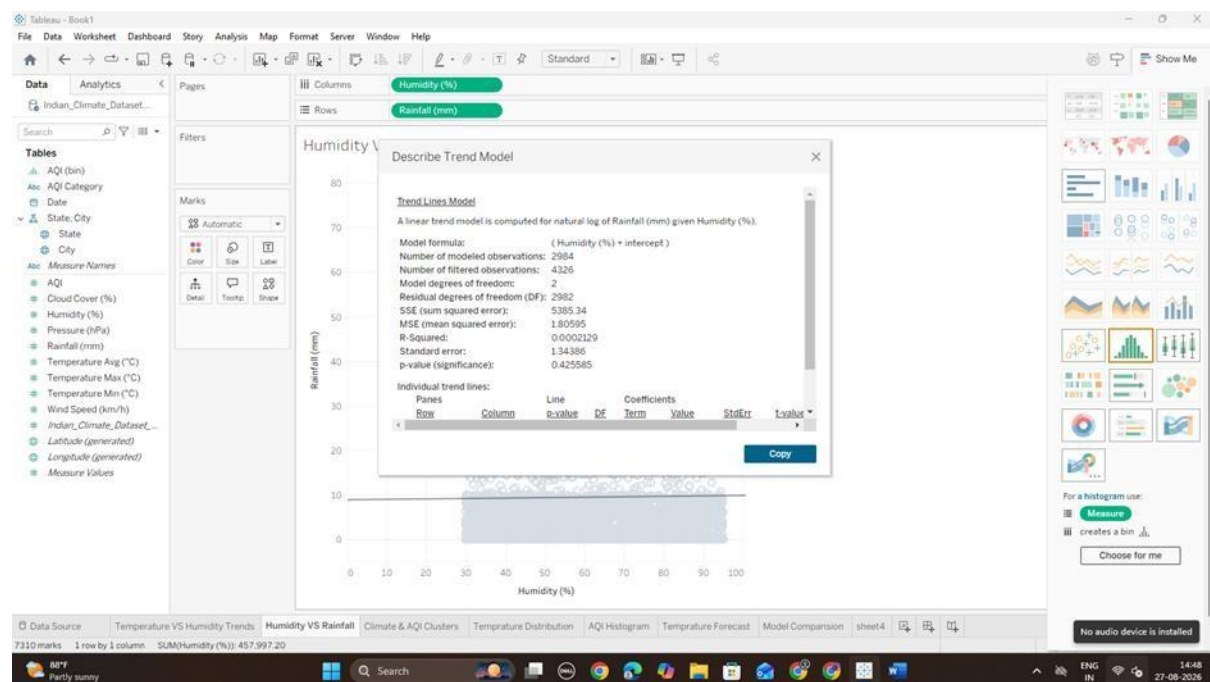
1. Create a worksheet named Model Comparison.
2. Drag **Temperature_Max (°C)** → **Columns**.
3. Drag **AQI** → **Rows**.
4. Drag **City** → **Detail**.
5. Open the Analytics pane.
6. Add a Linear Trend Line.
7. Record the R^2 and p-value.
8. Open Edit Trend Lines.
9. Compare Linear, Logarithmic, Exponential and Polynomial models.
10. Compare the R^2 values and identify the model with the better fit.

Output:

Steps:

1. Open the Model Comparison worksheet.
2. Right-click the trend line or chart.
3. Select Describe Trend Model.
4. For forecasting, select Describe Forecast.
5. Click Copy or Export to save the statistical results.
6. Record the Equation Parameters such as intercept and slope.
7. Record the R^2 value.
8. Record the p-value.
9. Interpret the statistical results and identify whether the relationship is significant.

Output:



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